

REDOX reactions

Example

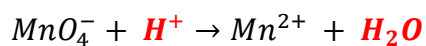
NaHSO_3 with KMnO_4

We forget about the Na, and K

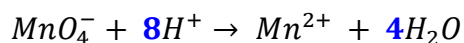
Two semi-reactions:

a) Reduction of oxidizing agent, gain electrons

we can say, it loses Oxygens = gains Hydrogens (H^+)



We balance it:

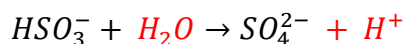


We do oxidation numbers, with Mn, at reactants, Mn^{+7} at products, Mn^{+2} , it gains 5e^-

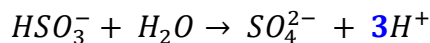


b) Oxidation of reducing agent, loses electrons:

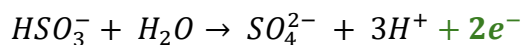
we can say, it gains oxygens, = loses Hydrogens (H^+)



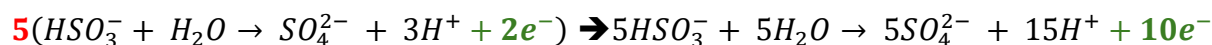
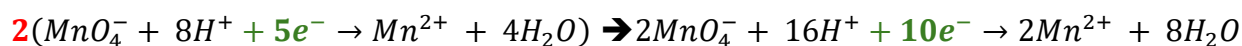
We balance it:



We do oxidation numbers, with S, at reactants, S^{+4} at products, S^{+6} , it loses 2e^-



Once we have the two semi-reactions, we treat them as algebraic equations ... we try to cancel the electrons:



We get net ionic equation:

